Tanmay Bhuskute

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SUMMARY

Backend Software Engineer and M.S. Computer Science candidate (Dec 2026) building backend services, cloud infrastructure, and agentic LLM systems using Java, Python, Go, AWS, and Kubernetes.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, Go, JavaScript, TypeScript, C/C++, SQL.

Backend & Web: Node.js, Django, Flask, FastAPI, Express, REST APIs, Microservices, React.

Databases: PostgreSQL, MySQL, Redis.

Cloud & DevOps: AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Jenkins, Linux, Bash, Azure, Google Cloud Platform.

AI & Agentic Systems: Gemini, Vertex AI, LangChain, MCP, PyTorch, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built agentic, two-stage LLM pipeline decoupling video understanding from device automation for Android bug reproduction, benchmarked against **ViBR** (GPT-4o baseline) across 53 runs.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** backend infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and cutting deployment time by **40%** across AWS cloud services.
- Debugged and resolved **44 critical production issues** in 4 months across Linux, EC2, databases, and Teamcenter X services spanning multiple levels of the stack.
- Designed **Ansible-based Blue-Green deployment** framework, cutting production upgrade downtime by **50%**; migrated **40+ environments** to Kubernetes via ArgoCD, Helm, and Vault.
- Served as Quality Coach, raising automated code coverage from **35% to 78%**; mentored **5 engineers** and cut monthly AWS spend **40%** ($15K to $9K) via EC2 rightsizing.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, FastAPI, PostgreSQL, PySpark, scikit-learn] Jul 2026

- Built backend ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset); served real-time predictions via a FastAPI REST API backed by PostgreSQL and Redis caching.
- Designed two-stage Poisson hurdle model improving forecast MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest, containerized with Docker for cloud deployment.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).